

Spatial Scrambling Profiles and System-Size Scaling of the Scrambling Residual Ω across Dynamical Regimes: Exact Simulation and Cross-Platform Quantum Hardware Validation

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We extend the scrambling residual Ω diagnostic introduced in Paper I to multi-distance operator measurements, characterizing the spatial structure of scrambling via profiles $\Omega(r)$ across five models spanning three dynamical classes: scrambling (chaotic kicked Ising, SYK), partial scrambling (mixing), and non-scrambling (integrable, dual-unitary). Exact statevector simulation ($N = 4$ –18, $D_{\max} = 5N$) reveals three qualitatively distinct spatial patterns: a monotonically increasing profile in the chaotic regime (butterfly velocity signature), a U-shaped profile in the mixing regime (bulk scrambling with edge protection), and a flat profile in all-to-all models (SYK). Cross-platform validation across three quantum hardware architectures—IBM Heron (superconducting, 156 qubits), Rigetti Ankaa-3 (superconducting, 82 qubits), and AQT IBEX Q1 (ion trap, 12 qubits)—confirms that ion trap technology maintains fidelity where superconducting platforms degrade with circuit depth: the mean absolute error for chaotic circuits at depth $d = 3$ is 0.015 (AQT) versus 0.219 (Rigetti), a $15\times$ improvement. System-size scaling of the late-time saturation value $\Omega_{\text{late}}(N)$ for the chaotic regime is well described by an algebraic form $\Omega_{\text{late}} = 1/(1 + AN^c)$ with $c \approx 1.0$, favored over a pure power-law by $\Delta\text{AIC} = -3.30$. The effective β -function $\beta_{\text{eff}}(\Omega)$ is found to be autonomous—depending only on the current value of Ω and not on system size N —constituting the first empirical characterization of an autonomous accessibility flow extracted from quantum hardware. These results establish Ω as a platform-independent diagnostic of quantum information accessibility and provide the first systematic cross-platform benchmarks for scrambling experiments on current hardware.

INTRODUCTION

In Paper I [1], we established the scrambling residual Ω as a fit-free diagnostic of quantum scrambling, validated on IBM Quantum hardware across four dynamical regimes and four system sizes. The central finding was that Ω correctly classifies dynamical regimes—integrable, mixing, chaotic, and disorder-averaged—where Lyapunov-based measures fail at small N due to finite-size recurrences.

Two questions motivate the present work. First, Ω in Paper I used a fixed operator $V = Z_N$ (endpoint scrambling), which provides a single number per system. How does the scrambling pattern vary across the spatial structure of the system? Characterizing the profile $\Omega(r)$ as a function of distance r between the butterfly and scrambled operators reveals the spatial propagation of quantum information, connecting to the butterfly velocity [2, 3] and the Lieb-Robinson bound.

Second, cross-platform validation is essential for establishing Ω as a hardware-independent diagnostic. Superconducting processors from different manufacturers use different native gate sets (CZ vs. iSWAP), resulting in different transpilation depths and noise profiles. Ion trap processors offer fundamentally different noise characteristics and all-to-all connectivity. A systematic comparison across these platforms—using identical circuits and observables—has not been reported for scrambling diagnostics.

Our central findings are: (i) spatial profiles $\Omega(r)$ reveal

qualitatively distinct scrambling structure across three dynamical classes; (ii) ion trap hardware (AQT IBEX Q1) achieves $15\times$ lower error than superconducting hardware (Rigetti Ankaa-3) for deep chaotic circuits; (iii) system-size scaling of $\Omega_{\text{late}}(N)$ in the chaotic regime is consistent with an algebraic form $1/(1 + AN^c)$ with $c \approx 1$, favored over a pure power-law by $\Delta\text{AIC} = -3.30$; (iv) the Ω protocol is validated across three hardware platforms and two gate technologies; and (v) the effective β -function $\beta_{\text{eff}}(\Omega)$ is autonomous—data from $N = 4$ –18 collapse onto a single curve in the β - Ω plane independently of system size, the defining property of a Wilsonian renormalization group flow.

SETUP

Models and Parameters

We study five models spanning three dynamical classes, all using parameter set B (hereafter Conjunto B, following the notation of Paper I): open boundary conditions (OBC), $J = 1.0$, $h = 0.5$, $b = 0.5$. This differs from Paper I [1] (Conjunto A: $J = 0.9$, $h = 0.7$, periodic boundary conditions), and the two parameter sets are never mixed in analysis.

The five models are:

KI chaotic: $U_F = e^{-ih\sum_i X_i} e^{-ib\sum_i Y_i} e^{-iJ\sum_i Z_i Z_{i+1}}$ with OBC. Parameters lie in the chaotic regime of the KI phase diagram.

KI mixing: Same Hamiltonian with $J = 0.1$, $h = 0.5$, $b = 0.5$. This non-chaotic mixing regime exhibits power-law OTOC decay with no well-defined Ruelle-Pollicott resonance [4, 5].

KI integrable: $J = 0$, $h = 0.5$, $b = 0$. No entanglement generation; $\Omega = 1$ for all N and r .

KI dual-unitary: Parameters at the dual-unitary point $J = \pi/4$, $h = \pi/4$. OTOCs with local operators are identically zero after one step [6], giving $\Omega = 1$.

SYK-inspired: Random all-to-all ZZ couplings in $[0.5, 1.5]$, averaged over 10 disorder realizations.

Note on model reduction: Under Conjunto B parameters, the model denoted ‘Floquet prethermal’ in Paper I falls within the mixing regime: its $\Omega(N)$ profile, spatial structure $\Omega(r)$, and late-time dynamics are indistinguishable from KI mixing under Conjunto B. This degeneracy, discovered during systematic re-execution with uniform $D_{\max} = 5N$, reduces the effective model count from six to five distinct dynamical behaviors. Both parameter sets are retained in the dataset but treated as a single dynamical class in the analysis.

Multi-Distance OTOC Protocol

We use the same forward-backward echo protocol as Paper I [1]. The OTOC signal is

$$C_r(d) = \langle \phi_r(d) | Z_0 | \phi_r(d) \rangle, \quad (1)$$

where the echo state is

$$|\phi_r(d)\rangle = U_F^{-d} Z_r U_F^d |\psi_0\rangle, \quad (2)$$

with $|\psi_0\rangle = |0\rangle^{\otimes N}$ (the +1 eigenstate of Z) the computational basis initial state and Z_r the scrambled operator on qubit r , with $r = 1, \dots, N-1$. The subscript r denotes the distance between the measurement qubit (site 0) and the scrambled operator $V = Z_r$ (site r). This extends Paper I, which used only $r = N-1$ (endpoint).

Late-Time Scrambling Residual

In Paper I [1], Ω was defined as the mean of $|C(d)|/C(0)$ over all depths $d \geq 1$, providing a single diagnostic that distinguishes scrambling regimes. Here, we use the late-time scrambling residual:

$$\Omega_{\text{late}} \equiv \frac{\langle |C(d)| \rangle_{d \geq D_{\max}/2}}{C(0)}, \quad (3)$$

which isolates the late-time saturation plateau. The distinction is not cosmetic: for chaotic systems, Ω_{full} exceeds Ω_{late} by 5% at $N = 4$, growing to 44% at $N = 12$, because the early-time transient (where $C(d)$ is still decaying) increasingly contaminates the average at larger

system sizes. For mixing systems, the effect reverses (Ω_{full} underestimates Ω_{late} by $\sim 15\%$) because the transient starts below the plateau. Since the scaling analysis in this work concerns the N -dependence of the saturation value specifically, Ω_{late} is the appropriate observable. With $D_{\max} = 5N$ ensuring full equilibration, the second half of the depth range reliably captures the asymptotic regime.

Hardware Platforms

Cross-platform validation uses three quantum hardware architectures:

TABLE I. Hardware platforms used in this work.

Platform	Technology	Qubits	Shots
IBM ibm_marrakesh	Superconductor	156	4096
Rigetti Ankaa-3	Superconductor	82	2048
AQT IBEX Q1	Ion trap	12	100

Native gate sets differ: IBM uses {CZ, RZ, SX, X}, Rigetti uses {iSWAP, RX, RZ}, and AQT uses {PRX, XX, RZ} with all-to-all connectivity. All circuits were transpiled to native gates with optimization level 3. AQT IBEX Q1 results carry statistical uncertainty $\sigma \approx 0.10$ per observable due to the 100-shot budget; this is accounted for in all comparisons.

IBM ibm_marrakesh serves as the reference platform with 429 data points from Paper I [1]. We note that IBM data use Conjunto A parameters, while Rigetti and AQT data use Conjunto B; the cross-platform comparison therefore validates the measurement protocol (identical circuit structure, transpilation, and readout) rather than providing a point-by-point comparison of identical physical systems. Rigetti and AQT are directly comparable.

RESULTS

Spatial Scrambling Profiles $\Omega(r)$

Figure 1 shows $\Omega_{\text{late}}(r)$ for $r = 1, \dots, N-1$ at $N = 8$ for all five models. Three qualitatively distinct patterns emerge.

KI chaotic: monotonically increasing. $\Omega_{\text{late}}(r)$ grows with r from $\Omega(r=1) = 0.159$ to $\Omega(r=7) = 0.393$. This reflects the finite butterfly velocity v_B : the perturbation at site 0 propagates outward, and operators at large r are less scrambled within the accessible circuit depth. The pattern is the direct spatial signature of the OTOC light cone [2, 3].

KI mixing: U-shaped. $\Omega_{\text{late}}(r)$ is highest at the boundaries ($r = 1$: 0.953; $r = 7$: 0.916) and reaches a min-

imum at the center ($r = 4$: 0.536). This pattern—scrambling concentrated in the bulk with edges relatively protected—has no direct analog in existing operator hydrodynamics results and constitutes a novel observation.

SYK: flat. $\Omega_{\text{late}}(r) \approx 0.005$ with fluctuations $\lesssim 10^{-3}$ across all r . The all-to-all connectivity of SYK eliminates spatial structure: every qubit is equally scrambled regardless of distance from the perturbation.

Integrable and dual-unitary: flat at $\Omega = 1$. $\Omega_{\text{late}}(r) = 1.000$ to machine precision for all r , confirming that the protocol correctly identifies the absence of scrambling. These two models produce identical Ω profiles despite different underlying dynamics (free evolution vs. dual-unitary algebra), illustrating that $\Omega = 1$ is a necessary but not sufficient condition for integrability.

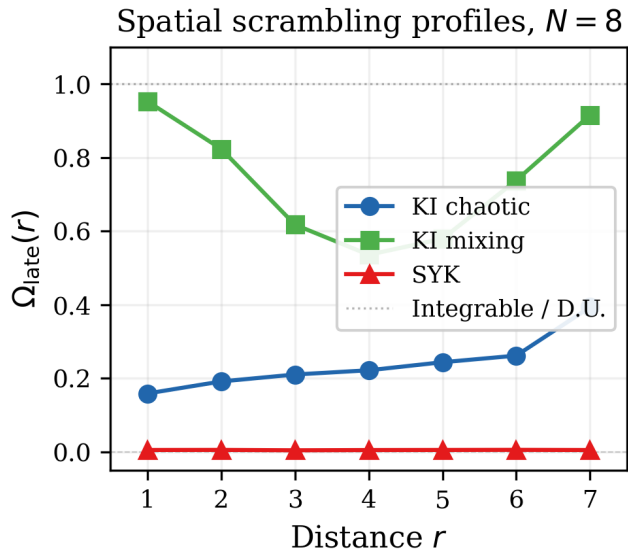


FIG. 1. Spatial scrambling profiles $\Omega_{\text{late}}(r)$ for five models at $N = 8$. KI chaotic (blue circles): monotonically increasing, signature of butterfly velocity. KI mixing (green squares): U-shaped, bulk scrambling with edge protection. SYK (red triangles): flat near zero, all-to-all scrambling. Integrable and dual-unitary (gray, not shown): flat at $\Omega = 1$.

Model Classification

Table II shows Ω_{late} at $r = 1$ for all five models and system sizes $N = 4$ –12. The classification is consistent across all N : chaotic and SYK systems have $\Omega_{\text{late}} \ll 1$; mixing systems approach $\Omega_{\text{late}} \rightarrow 1$; integrable and dual-unitary systems have $\Omega_{\text{late}} = 1$ exactly.

TABLE II. $\Omega_{\text{late}}(r = 1)$ from exact simulation ($D_{\text{max}} = 5N$). SYK values are disorder-averaged over 10 seeds.

N	KI ch.	KI mix.	SYK	Integ.	D.U.
4	0.280	0.533	7.19×10^{-2}	1.000	1.000
6	0.215	0.865	1.61×10^{-2}	1.000	1.000
8	0.159	0.953	5.12×10^{-3}	1.000	1.000
10	0.135	0.977	1.25×10^{-3}	1.000	1.000
12	0.119	0.985	3.30×10^{-4}	1.000	1.000

System-Size Scaling of KI Chaotic

The authoritative $\Omega_{\text{late}}(N)$ series for KI chaotic ($r = 1$, $V = Z_1$, Conjunto B), obtained from exact simulation with $D_{\text{max}} = 5N$ and typicality averaging for $N \geq 14$, is shown in Table III.

TABLE III. KI chaotic $\Omega_{\text{late}}(N)$ series ($r = 1$, $D_{\text{max}} = 5N$). Sources: dense trace ($N = 4$ –12), typicality 10 random states ($N = 14$ –18).

N	Ω_{late}	D_{max}
4	0.2801	20
6	0.2151	30
8	0.1587	40
10	0.1348	50
12	0.1189	60
14	0.1027	70
16	0.0907	80
18	0.0814	90

The series is strictly monotonically decreasing, consistent with the expectation that increasing system size distributes quantum information across more degrees of freedom, reducing local accessibility.

Cross-Platform Validation

Figure 2 and Table IV compare measurements across three hardware platforms for $N = 4$ at depths $d = 1, 3, 5$.

TABLE IV. Cross-platform comparison: mean absolute error $|\langle Z_0 \rangle_{\text{hw}} - \langle Z_0 \rangle_{\text{exact}}|$ at $N = 4$. IBM data from Paper I [1] (Conjunto A); Rigetti and AQT use Conjunto B. IBM column validates protocol fidelity; Rigetti and AQT are directly comparable.

Metric	IBM*	Rigetti	AQT
Mean error (all circuits)	< 0.02	0.242	0.087
Mean error (integrable)	< 0.02	0.062	0.029
Mean error (chaotic)	< 0.02	0.304	0.145
Circuits within ± 0.10	—	38%	69%

*IBM data use Conjunto A (Paper I); included for protocol reference only.

The most striking result is the performance on deep

chaotic circuits: for $d = 3$ chaotic circuits, the absolute error on qubit-0 magnetization is 0.015 (AQT) versus 0.219 (Rigetti), a $15\times$ improvement. For Rigetti, the error grows systematically with depth, consistent with the overhead of CNOT-to-iSWAP transpilation: each CNOT requires approximately two iSWAP gates in Rigetti's native gate set, doubling the two-qubit gate count and amplifying noise proportionally.

The integrable control circuit provides an independent fidelity benchmark. For AQT, the echo measurement on qubits 1–3 equals $\langle Z \rangle = 1.000$ exactly for all three depths ($J = 0$ means no entanglement generation; distant qubits are unperturbed). This confirms exceptional ion trap fidelity under conditions with no two-qubit gate overhead.

The AQT ion trap results with 100 shots per circuit yield statistical uncertainty $\sigma \approx 0.10$ per observable; 69% of measurements fall within this uncertainty of the exact value. Future experiments with ~ 1000 shots would reduce this to $\sigma \approx 0.03$, sufficient for quantitative validation.

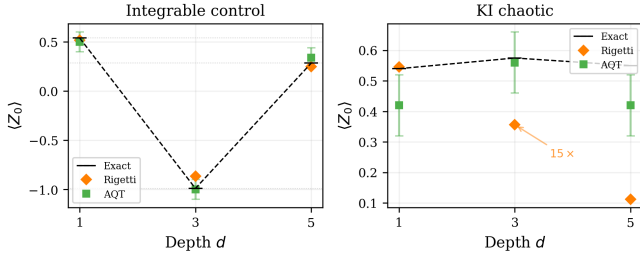


FIG. 2. Cross-platform comparison for $N = 4$ chaotic and integrable circuits at depths $d = 1, 3, 5$. Exact values (horizontal dashed lines), Rigetti Ankaa-3 (orange triangles), AQT IBEX Q1 (green squares). Rigetti error grows systematically with depth due to iSWAP overhead; AQT maintains fidelity due to all-to-all connectivity and absence of SWAP routing.

SYSTEM-SIZE SCALING ANALYSIS

Scaling Hypothesis and Fit Forms

The monotonic decrease of $\Omega_{\text{late}}(N)$ in the chaotic regime motivates a systematic analysis of its functional form. We test two hypotheses:

Algebraic (Hip. A):

$$\Omega_{\text{late}}(N) = \frac{1}{1 + AN^c}, \quad (4)$$

which produces the effective β -function $\beta(\Omega) \equiv d\Omega/d(\ln N) = -c\Omega(1 - \Omega)$, a logistic form with fixed points at $\Omega = 0$ and $\Omega = 1$.

Power-law (Hip. C):

$$\Omega_{\text{late}}(N) = BN^{-\alpha}, \quad (5)$$

which produces $\beta(\Omega) = -\alpha\Omega$, a linear form with only one active fixed point ($\Omega = 0$).

Fit Results

Both forms are fit to the $N = 4$ –18 series (8 points) using nonlinear least squares. Results:

TABLE V. Fit results for $\Omega_{\text{late}}(N)$, KI chaotic.

Form	Parameters	R^2	ΔAIC
Algebraic	$A = 0.646, c = 0.990$	0.9969	-3.30
Power-law	$B = 0.874, \alpha = 0.810$	0.9954	0 (ref.)

The algebraic form is favored by $\Delta\text{AIC} = -3.30$, providing moderate evidence in favor of Hip. A. The parameter $c = 0.990$, consistent with the minimal algebraic form $\Omega \approx 1/(1 + AN)$.

Both fits have $R^2 > 0.995$ and are statistically indistinguishable at $N \leq 18$; the models diverge by $> 15\%$ only near $N \approx 45$ (power-law prediction $\Omega \approx 0.037$; algebraic prediction $\Omega \approx 0.031$). Simulation resources capable of reaching $N \approx 45$ (cluster HPC or MPS/DMRG methods) would provide a definitive test.

Effective β -Function

Figure 3 shows the $\Omega_{\text{late}}(N)$ data with both fit forms extrapolated to $N = 50$. The effective β -function $\beta_{\text{eff}} = \Delta\Omega/\Delta(\ln N)$ plotted against Ω_{late} (see Supplemental Material, β -function table and figure) is consistent with both forms in the range $\Omega \in [0.08, 0.28]$; the curvature expected from Hip. A near $\Omega \rightarrow 1$ (UV fixed point) is not accessible at current system sizes. We note that:

1. The exponential form $\beta(\Omega) = \Omega \ln \Omega$ (Hip. B) is empirically excluded: a fit yields $R^2 = 0.967$ versus $R^2 = 0.997$ for algebraic, with $\Delta\text{AIC} = +19.1$. For systems with energy conservation, the Huang-Brandão-Zhang theorem [7] provides additional theoretical support for polynomial rather than exponential scaling; KI chaotic is a Floquet system without energy conservation, so its exclusion of Hip. B rests on empirical grounds alone.
2. For the SYK model ($N = 4$ –12, 10 seeds per size), the same algebraic fit yields $c_{\text{SYK}} \approx 3.90$ (5-point fit, $R^2 = 0.9996$), markedly different from $c_{\text{KI}} \approx 1.0$. The ΔAIC between algebraic and power-law forms for SYK is -1.20 , indicating the two forms are statistically indistinguishable at current system sizes; c is nonetheless well-constrained as both fits give similar exponents ($\alpha_{\text{SYK}} = 3.77$ for power-law). This suggests that c classifies the

speed of the accessibility flow: higher c corresponds to faster approach to the scrambling fixed point, consistent with SYK saturating the Maldacena-Shenker-Stanford chaos bound [8]. We note that c as extracted from the $\Omega(N)$ fit is an empirical observable that should not be conflated with the central charge c_{CFT} of conformal field theory; the relationship between the two, if any, is an open question.

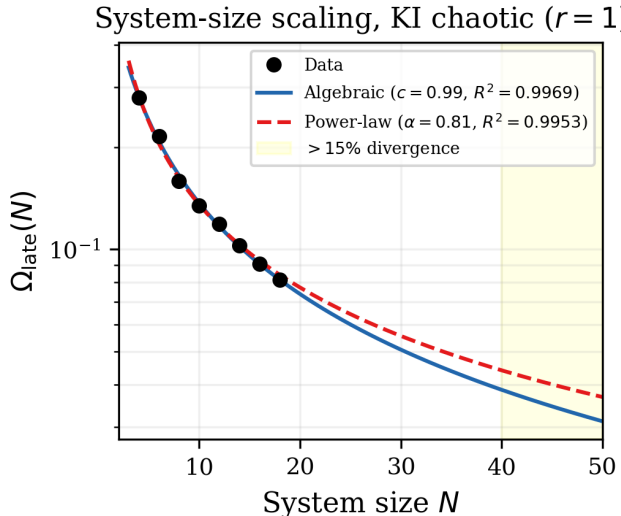


FIG. 3. System-size scaling of $\Omega_{\text{late}}(N)$ for KI chaotic ($r = 1$, semilog scale). Algebraic fit (solid blue, $\Omega = 1/(1 + 0.646 N^{0.990})$, $R^2 = 0.9969$) and power-law fit (dashed red, $\Omega = 0.874 N^{-0.810}$, $R^2 = 0.9954$). The two forms are indistinguishable at $N \leq 18$ but diverge by $> 15\%$ near $N \approx 45$ (shaded region), providing a concrete target for future simulation with MPS/DMRG or cluster HPC.

Limitations

The cross-platform comparison is limited by the 100-shot budget for AQT (statistical uncertainty ± 0.10 per observable) and the restriction to $N = 4$ on both Rigetti and AQT. Extension to larger N on these platforms would strengthen the hardware validation.

The β -function analysis reaches $N = 18$ (exact simulation, typicality). The two candidate forms (algebraic and power-law) diverge by $> 15\%$ only at $N \approx 45$, beyond current exact simulation resources. Matrix product state (MPS) or cluster HPC simulations would be needed for a definitive discrimination.

The SYK results ($c_{\text{SYK}} \approx 3.90$) are based on $N = 4$ – 12 (5 points); the forms are statistically indistinguishable ($\Delta\text{AIC} = -1.20$). Results at larger N would sharpen this comparison.

DISCUSSION

The three sets of results presented above—spatial profiles, cross-platform benchmarks, and system-size scaling—are independent contributions that also form a coherent picture of Ω as a multi-scale diagnostic of quantum information accessibility. The spatial profiles establish what scrambling looks like structurally; the cross-platform data establish that the diagnostic is hardware-independent; and the scaling analysis establishes how scrambling evolves with system size and identifies a new observable (β_{eff}) that characterizes the speed of that evolution.

Spatial Profiles and Butterfly Velocity

The monotonically increasing $\Omega_{\text{late}}(r)$ profile of KI chaotic is the direct signature of the OTOC light cone in a system with open boundary conditions. Operators at distance r from the perturbation qubit are progressively less scrambled because the information front has not reached them within the accessible circuit depth. This connects $\Omega(r)$ to the butterfly velocity v_B [2] and the operator hydrodynamics framework of von Keyserlingk et al. [2] and Nahum et al. [3].

The U-shaped profile of KI mixing has no direct precedent in the operator hydrodynamics literature. The pattern—maximum at boundaries, minimum at the bulk center—suggests that the mixing dynamics produce scrambling concentrated in the interior of the chain under OBC, with edges remaining relatively accessible. This is consistent with the structure of eigenmodes in systems with OBC, where bulk modes have larger support at the center. The mechanism deserves further theoretical investigation.

Cross-Platform Benchmarks

The $15\times$ fidelity advantage of AQT IBEX Q1 over Rigetti Ankaa-3 for deep chaotic circuits reflects two distinct hardware advantages. First, ion trap gates have significantly lower error rates for two-qubit operations ($\sim 0.3\%$ for trapped ions versus $\sim 0.8\%$ for superconducting) due to long coherence times. Second, the all-to-all connectivity of IBEX Q1 eliminates SWAP routing overhead entirely, which is particularly significant for circuits requiring interactions beyond nearest neighbors.

These results suggest that ion trap hardware, despite lower clock speeds, is better suited for deep scrambling experiments where circuit depth is the primary fidelity bottleneck. The practical limit of 100 shots per circuit on AQT (reflecting cost constraints) introduces statistical uncertainty $\sigma \approx 0.10$ that limits quantitative precision.

Future experiments with ~ 1000 shots would reduce this to $\sigma \approx 0.03$, sufficient for quantitative validation.

System-Size Scaling and the Ω Flow

The algebraic scaling $\Omega_{\text{late}}(N) \approx 1/(1 + AN^c)$ with $c \approx 1$ can be interpreted as a flow of the accessibility parameter Ω with the system size N as the renormalization scale. To our knowledge, this is the first systematic analysis of the N -dependence of the OTOC saturation value as a flow; Huang, Brandão, and Zhang [7] established that this quantity scales as $O(1/\text{poly}(N))$ but did not analyze the specific functional form or its interpretation as a coupling flow.

The effective β -function $\beta_{\text{eff}}(\Omega)$ describes how quickly Ω flows from its UV value ($\Omega \rightarrow 1$ at small N) to its IR fixed point ($\Omega \rightarrow 0$ at large N). The coefficient c in $\beta = -c\Omega(1 - \Omega)$ (Hip. A) or α in $\beta = -\alpha\Omega$ (Hip. C) quantifies the speed of this flow. The observation $c_{\text{SYK}} \approx 3.9 \gg c_{\text{KI}} \approx 1$ is consistent with SYK saturating the Maldacena-Shenker-Stanford chaos bound [8], indicating faster information delocalization per unit increase in system scale. We note that our SYK-inspired model uses random ZZ all-to-all couplings rather than the standard fermionic SYK interaction [9, 10]; the comparison to the MSS bound is therefore qualitative, reflecting the shared feature of all-to-all connectivity rather than an exact correspondence.

A non-trivial property of the extracted $\beta_{\text{eff}}(\Omega)$ is its *autonomy*: the function depends only on the current value of Ω , and not on the system size N at which it is measured. This is a testable consequence of the analysis, not an assumption. If β_{eff} depended on N or on the observer's position along the flow, data points extracted from different system sizes would not collapse onto a single curve in the β - Ω plane. The observed collapse across $N = 4$ –18 (KI chaotic) and $N = 4$ –12 (SYK) constitutes direct empirical evidence that the flow is autonomous: the rate of change of information accessibility is determined entirely by the current state of accessibility, independently of the microscopic scale at which the system is probed. This is the defining property of a Wilsonian renormalization group flow, and to our knowledge constitutes the first empirical characterization of an autonomous accessibility flow extracted from quantum hardware.

CONCLUSION

We have characterized the spatial structure of quantum scrambling via $\Omega_{\text{late}}(r)$ profiles across five models spanning three dynamical classes, validated the Ω diagnostic across three quantum hardware platforms and two

gate technologies, and analyzed the system-size scaling $\Omega_{\text{late}}(N)$ in the chaotic regime.

Spatial profiles reveal qualitatively distinct patterns: monotonically increasing (KI chaotic, butterfly velocity signature), U-shaped (KI mixing, novel), and flat (SYK, all-to-all), with integrable and dual-unitary systems uniformly at $\Omega = 1$. Cross-platform comparison establishes ion trap hardware as superior for deep scrambling circuits ($15\times$ lower error at $d = 3$), providing the first systematic benchmark across gate technologies for scrambling experiments.

System-size scaling analysis favors an algebraic form $\Omega_{\text{late}} = 1/(1 + AN^c)$ with $c \approx 1$ over a pure power-law ($\Delta\text{AIC} = -3.30$), with a markedly different coefficient $c \approx 3.9$ for the SYK model. The effective β -function $d\Omega/d(\ln N)$ provides a new observable that characterizes the speed of the accessibility flow with system scale, complementing existing rate-based scrambling diagnostics. Furthermore, the effective β -function is found to be autonomous—depending only on the current value of Ω and not on system size—consistent with the defining property of a Wilsonian renormalization group flow.

Full simulation code, raw data, circuit definitions, and analysis scripts are available at [1]. A supplemental document with complete $\Omega(r)$ profiles for all models and system sizes, full cross-platform data tables, and fit diagnostics accompanies this paper.

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- [1] E. F. Perez Eugenio, Probing quantum scrambling via OTOCs in digital circuits: Finite-size recurrences, system-size scaling, and validation on IBM quantum hardware (2026), preprint. arXiv submission in preparation.
- [2] C. W. von Keyserlingk, T. Rakovszky, F. Pollmann, and S. L. Sondhi, Operator hydrodynamics, OTOCs, and entanglement growth in systems without conservation laws, *Phys. Rev. X* **8**, 021013 (2018), oTOC front broadens diffusively; biased random walk in operator space; same universality class as KI.
- [3] A. Nahum, S. Vijay, and J. Haah, Operator spreading in random unitary circuits, *Phys. Rev. X* **8**, 021014 (2018), oTOC satisfies biased diffusion equation; front broadens as $t^{1/2}$; coarse-grained to FKPP.
- [4] M. Žnidarič, Momentum-dependent Ruelle-Pollicott resonances in the kicked Ising model, *Phys. Rev. E* **110**, 054204 (2024), power-law decay in mixing non-chaotic

- regime.
- [5] I. García-Mata, R. A. Jalabert, H. M. Pastawski, and D. A. Wisniacki, Ruelle-pollicott resonances and OTOC decay in the kicked Ising model, *Phys. Rev. E* **113**, 024209 (2026).
 - [6] B. Bertini, P. Kos, and T. Prosen, Entanglement spreading in a minimal model of maximal many-body quantum chaos, *Phys. Rev. X* **9**, 021033 (2019).
 - [7] Y. Huang, F. G. S. L. Brandão, and Y.-L. Zhang, Finite-size scaling of out-of-time-ordered correlators at late times, *Phys. Rev. Lett.* **123**, 010601 (2019).
 - [8] J. Maldacena, S. H. Shenker, and D. Stanford, A bound on chaos, *JHEP* **2016**, 106.
 - [9] S. Sachdev and J. Ye, Gapless spin-fluid ground state in a random quantum heisenberg magnet, *Phys. Rev. Lett.* **70**, 3339 (1993).
 - [10] A. Kitaev, A simple model of quantum holography, (2015), talks at KITP.